
From Uniform Elevation Uncertainty to Gaussian Height Distribution in Sonar Mapping

A Self-Contained Mathematical Derivation

*PhD Research · Seabed-to-Sky LiDAR/Sonar Mapping
April 2026*

Abstract

In 2-D imaging sonar, the elevation angle ϕ of each acoustic return is unknown. This note derives, step-by-step, the probability distribution of the vertical coordinate z induced by a uniform prior on ϕ . Under a small-angle approximation the induced distribution on z is also uniform. When semantic constraints are available (e.g. seabed label from a classifier), they enter as a Gaussian likelihood. A Bayesian update of the uniform prior with this likelihood yields a Gaussian posterior whenever the constraint uncertainty is much smaller than the beam footprint. The same result is recovered independently via the Maximum-Entropy (MaxEnt) principle.

1. Introduction

Imaging sonar systems such as the Blueview or Oculus generate a two-dimensional fan-beam measurement: for every beam the sonar reports the slant range r and the azimuth angle θ , but provides no information about the elevation angle ϕ . Because of this fundamental ambiguity every sonar return is consistent with a continuum of 3-D positions that form an arc in the vertical plane.

The standard way to handle this uncertainty probabilistically is to assign a prior distribution to ϕ and then propagate it to the vertical coordinate z . The simplest physically motivated prior is a uniform distribution over the one-way beam half-angle $\phi_{\max}$. This note shows that, under a small-angle approximation, that uniform prior on ϕ induces a uniform distribution on z . It then shows two mathematically independent routes by which semantic information converts that uniform distribution into a Gaussian: (i) a Bayesian update with a Gaussian likelihood, and (ii) the Maximum-Entropy principle with moment constraints.

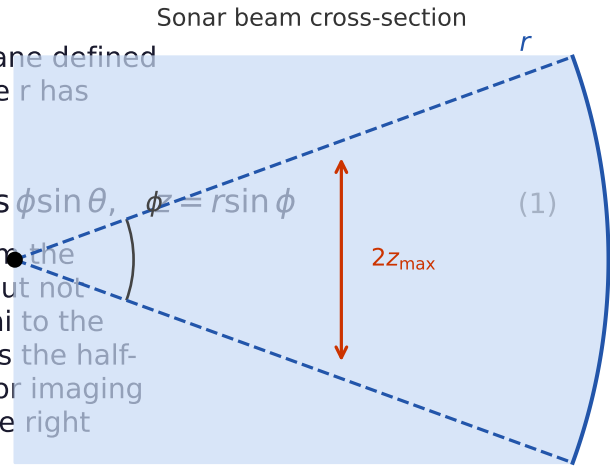
All equations are numbered and cited to standard references. The derivation is self-contained; only elementary probability calculus is assumed.

2. The Sonar Elevation Ambiguity

Consider a sonar that operates in a vertical plane defined by azimuth angle θ . A return at slant range r has Cartesian coordinates:

$$x = r \cos \phi \cos \theta, \quad y = r \cos \phi \sin \theta, \quad z = r \sin \phi \quad (1)$$

where ϕ is the elevation angle measured from the horizontal plane. The sonar reports (r, θ) but not ϕ . The one-way beam aperture constrains ϕ to the interval $[-\phi_{\max}, \phi_{\max}]$, where $\phi_{\max}$ is the half-angle of the beam (typically 5 deg to 20 deg for imaging sonars [Blueview, Oculus]). The diagram on the right illustrates the resulting arc of uncertainty.



3. Step 1 — Uniform Prior on the Elevation Angle

In the absence of any other information, every elevation angle within the beam aperture is equally likely. We therefore assign the non-informative (maximum-entropy) prior [1, Sec. 2.4]:

$$f_{\phi}(\phi) = \frac{1}{2\phi_{\max}}, \quad \phi \in [-\phi_{\max}, \phi_{\max}] \quad (2)$$

and $f_{\phi}(\phi) = 0$ otherwise. This is the unique distribution that is invariant to any relabelling of angles within the beam — i.e., it encodes complete ignorance about elevation [2, p. 46].

Remark: equation (2) treats the beam as having a hard boundary. A soft apodization window (e.g., a raised cosine) would change the shape of the prior, but the key structural results below remain the same.